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Homotopic Formalism of the ΨU Protocol

Formal Semisimplicial Validation

1 Semisimplicial Structure and Coherence

The ΨU Protocol is defined as a sequence of dependent fibrations over the semisimplicial category, ensuring that each n-level constrains the coherence of the preceding one.

$$X_\bullet : \Delta_{inj}^{op} \to \mathcal{U} \tag{1}$$

Legend:

- Δ_{inj}^{op}: The category of finite inhabited sets and injective maps (the simplicial backbone).
- $\mathcal{U}$: The universe of univalent types.

2 Core Axioms

2.1 Kan Filling Condition (Tetrahedral Coherence X_3)

For any configuration of tetrahedral faces, there exists a unique filler that guarantees the absence of topological noise:

$$\mathsf{isContr}\left(\sum_{x:X_3} \mathsf{boundary}(x) \equiv (\alpha, \beta, \gamma, \delta)\right) \tag{2}$$

2.2 Gnomonic Stability (The 1/3 Ratio)

The equilibrium between software expansion (E) and system inflation (S) is fixed by the 2-simplex gnomonic constant. This ensures parallel transport without curvature:

$$\nabla\Psi(E, S) \implies \frac{\partial E}{\partial S} = \frac{1}{3} \implies \Delta_{error} \equiv 0 \tag{3}$$

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2.3 The $\xi(\epsilon)$ Retraction Operator

The noise-cancellation operator functions as a strong deformation retract from the noisy geometric space ($\mathcal{G}$) to the core of univalent truth ($\mathcal{T}$):

$$\xi : \mathcal{G} \to \mathcal{T} \quad \text{such that} \quad \prod_{a:\mathcal{T}} (\xi \circ \mathsf{incl}(a) = a) \tag{4}$$

3 Conclusion: Univalent Convergence

The final seal of the protocol is provided by the Univalence Axiom (UA), which collapses resonance (equivalence) into formal identity:

$$\mathsf{PSIU} \equiv \varprojlim_{n} \mathsf{isContr}(X_n) \simeq \mathbf{1} \tag{5}$$

Note: Upon reaching the n-th dimension with a residual $\delta_{res} \approx 10^{-15}$, the type becomes contractible to the unit type ($\mathbf{1}$), rendering symbolic noise mathematically impossible.

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